

TechArena 2025 - Phase 1 Submission Instructions

Last Updated: 26-08-2025

Overview

Welcome to TechArena 2025! This document provides comprehensive instructions for submitting your Phase 1 solution. Please read all sections carefully to ensure your submission is properly validated and evaluated.

Submission Requirements

Required Deliverables

Your submission must include **all** of the following components:

1. **Main Script:** `main.py` (in project root directory)
2. **Input Data Processing:** Code to read and process the provided Excel input file
3. **Output Files:** Three result files in CSV or Excel format
4. **Dependencies:** `requirements.txt` file (if using external packages)
5. **Documentation:** `README.md` file explaining your approach

Output Files Required

Your `main.py` script must generate exactly **three output files** with the following names (CSV or Excel format):

1. **TechArena_Phase1_Configuration.csv** (or `.xlsx`)
 - Optimal BESS configuration parameters
 - Required columns: C-rate, number of cycles, yearly profits [kEUR/MW], levelized ROI [%]
2. **TechArena_Phase1_Investment.csv** (or `.xlsx`)
 - Investment analysis and ROI calculations
 - Must include: WACC, inflation rate, discount rate, yearly profits, year-by-year analysis, levelized ROI
3. **TechArena_Phase1_Operation.csv** (or `.xlsx`)
 - Required columns: Timestamp, Stored energy [MWh], SoC [-], Charge [MWh], Discharge [MWh], Day-ahead buy [MWh], Day-ahead sell [MWh], FCR Capacity [MW], aFRR Capacity POS [MW], aFRR Capacity NEG [MW]

Project Structure

Your submission should follow this **exact** directory structure:

```

1 your_team_submission.zip
2 main.py                # Main execution script (REQUIRED)
3 requirements.txt        # Python dependencies (if needed)
4 README.md              # Documentation (RECOMMENDED)
5 input/                 # Input data directory
6   TechArena2025_ElectricityPriceData_v2.xlsx
7 [additional files/folders] # Your implementation files
8   output/              # Output data directory

```

Listing 1: Required Directory Structure

Critical Requirements

- **main.py** **MUST** be in the **root directory** (not in subdirectories)
- Input file **MUST** be in **input/** subdirectory
- Output files will be generated in the output directory
- **No absolute file paths** - use relative paths only

Technical Specifications

Python Requirements

- **Python Version:** 3.8 or higher
- **Output Format:** CSV (.csv) or Excel (.xlsx) files

Input Data Structure

The input file contains 4 sheets:

- **Day-ahead prices:** Historical electricity prices for 2024
- **FCR prices:** Frequency Containment Reserve prices
- **aFRR capacity prices:** Automatic Frequency Restoration Reserve prices
- **Data description:** Market conditions and BESS parameters

Data Validation Constraints

Configuration File:

- Must include required parameters: yearly profits [kEUR/MW], levelized ROI [%] for all C-rate and number of cycles combinations

Investment File:

- Must include required parameters: WACC, inflation rate, discount rate
- Must include Levelized ROI calculation

Parameter	Value
WACC	value
Inflation Rate	value
Discount rate	value
Yearly profits (2024)	value

Year	Initial Investment [kEUR/MWh]	Yearly profits [kEUR/MWh]
2023		
2024		
2025		
2026		
2027		
2028		
2029		
2030		
2031		
2032		
2033		
Levelized ROI		Value

- Below is the sample table for reference:

Operation File:

- Must include required parameters: Timestamp, Stored energy [MWh], SoC [-], Charge [MWh], Discharge [MWh], Day-ahead buy [MWh], Day-ahead sell [MWh], FCR Capacity [MW], aFRR Capacity POS [MW], aFRR Capacity NEG [MW]

Implementation Guidelines

Example main.py Structure

```

1 import os
2 import pandas as pd
3
4 def main():
5     """
6     Main execution function for TechArena Phase 1 solution
7     """
8     # Read input data
9     input_file = os.path.join("input", "TechArena2025_ElectricityPriceData.xlsx")
10
11     # Process data and run optimization
12     # ... your implementation here ...
13
14     # Generate output files
15     config_df.to_csv("TechArena_Phase1_Configuration.csv", index=False)
16     investment_df.to_csv("TechArena_Phase1_Investment.csv", index=False)
17     operation_df.to_csv("TechArena_Phase1_Operation.csv", index=False)
18
19     print(" All output files generated successfully!")
20
21 if __name__ == "__main__":
22     main()

```

Listing 2: Example main.py Structure

Best Practices

DO:

- Use relative paths for all file operations
- Check if input files exist before processing
- Handle exceptions gracefully with try-catch blocks
- Validate your output before submission
- Test your script in a clean environment
- Use meaningful variable names and add comments
- Include all required dependencies in requirements.txt

DON'T:

- Use absolute file paths (e.g., `C:\Users\YourName\...`)
- Hardcode team-specific paths or credentials
- Rely on files not provided in the input
- Create infinite loops or excessive processing
- Leave debugging print statements that flood output
- Submit incomplete or placeholder files

Sample requirements.txt

```
1 pandas>=1.5.0
2 numpy>=1.21.0
3 openpyxl>=3.0.9
4 matplotlib>=3.5.0
5 scipy>=1.8.0
```

Listing 3: Sample requirements.txt

Submission Process

Step 1: Prepare Your Submission

1. Complete your implementation in `main.py`
2. Test thoroughly with the provided input data
3. Verify all output files are generated correctly
4. Create/update `requirements.txt` with all dependencies
5. Write `README.md` explaining your approach and any special instructions

Step 2: Test Your Submission

Before submitting, test your solution:

```
1 # Create a clean test environment
2 cd your_project_directory
3
4 # Install dependencies (if using virtual environment)
5 pip install -r requirements.txt
6
7 # Run your main script
8 python main.py
9
10 # Verify output files are created
11 ls -la *.csv # or *.xlsx
```

Listing 4: Testing Commands

Step 3: Create Submission Package

1. Create a ZIP file containing your entire project
2. Name the file: `TeamName_TechArena2025_Phase1.zip`
3. Verify contents match the required structure
4. Test the ZIP file by extracting and running in a new location

Step 4: Submit

1. Upload your ZIP file to the designated submission platform
2. Confirm successful submission before the deadline

Automated Validation

Your submission will undergo **automated validation** that checks:

Structure Validation

- Correct file and directory structure
- Presence of required files (`main.py`, output files)
- Proper file naming conventions

Code Quality

- Script executability and syntax
- Import statement validation
- Function and documentation presence
- Excel/CSV handling capabilities

Execution Testing

- Virtual environment setup and dependency installation
- Error handling and output generation

Output Validation

- Correct number and format of output files
- Schema compliance (column names and types)
- File size and content reasonableness

Common Issues and Solutions

Issue: “main.py not found in project root”

Solution: Ensure `main.py` is directly in the ZIP file root, not in a subdirectory.

Issue: “Input file not found”

Solution: Place input file in `input/` subdirectory and use relative path:

```
input_file = os.path.join("input", "TechArena2025_ElectricityPriceData_v2.xlsx")
```

Issue: “Requirements installation failed”

Solution:

- Test your `requirements.txt` in a clean environment
- Use specific version numbers for critical packages
- Include `openpyxl` for Excel file handling

Issue: “Output file validation failed”

Solution:

- Check column names match exactly (case-sensitive)
- Verify data types and ranges
- Ensure no missing values in required fields

Issue: “Script execution timeout”

Solution:

- Optimize your algorithms for efficiency
- Avoid unnecessary data loading or processing
- Consider using vectorized operations with `pandas/numpy`

Support and Contact

Technical Support

- **Email:** amandine@bemyapp.com

Only For Code Submission Support

- **Email:** mangesh.mankar@huawei.com

FAQ and Updates

- **Event Website:** <https://techarena-nuremberg.hackathon.com/>
- **Discussion Forum:** Available on the event platform
- **Live Q&A Sessions:** Scheduled times will be announced

Submission Checklist

Before submitting, verify:

- ☐ `main.py` is in the root directory
- ☐ Input file is in `input/` subdirectory
- ☐ Script runs successfully and generates all 3 output files
- ☐ `requirements.txt` includes all dependencies
- ☐ `README.md` explains your approach
- ☐ ZIP file is named correctly: `TeamName_TechArena2025_Phase1.zip`
- ☐ No absolute file paths or hardcoded paths
- ☐ All output files have correct column names
- ☐ Tested in a clean environment

Good luck, and may the best solution win!

TechArena 2025 Organizing Committee
Last Updated: 26-08-2025